

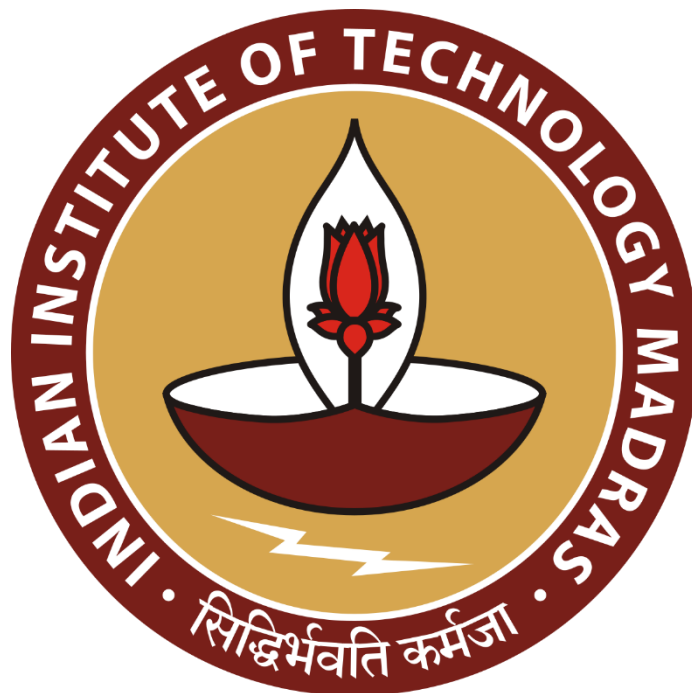
Data-Driven Strategies for Optimizing Profitability and Operations in a Saree Weaving Business

A Final Report for the BDM Capstone Project

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Contents

1. Executive Summary	2
2. Detailed Explanation of Analysis Process/Method	
2.1 Data Cleaning and Preprocessing	3
2.2 Comprehensive Explanation of Methods and Analysis	4-8
2.3 Justification of Tools and Methods	8
3. Results and Findings	
3.1 Design-wise Sales Contribution and Product Mix	9
3.2 Raw Material Consumption and Production Capacity Analysis	9
3.3 Cost Structure Analysis (Aara and Lachha Sarees)	9
3.4 Month-wise Profit and Loss Trends	10
3.5 Break-Even Analysis Results	10
3.6 Time Series Decomposition of Weekly Sales Data based on Designs	11-13
3.7. Saree Colour Shade Sales Analysis	14-15
4. Interpretation of Results and Recommendations	
4.1 Interpretation of Results	16
4.2 Actionable Recommendations	17-18
4.3. Impact and Benefits of Implementation	18

1. Executive Summary

Vishwas Maurya Textile is a small-scale powerloom manufacturing unit engaged in producing two primary saree designs—Aara and Lachha—for dealers and retailers. Despite stable demand and continuous operations, the business has been experiencing unclear profitability, inconsistent pricing decisions, and a lack of structured cost monitoring. Production and pricing decisions were primarily driven by sales volume rather than contribution margin or break-even analysis, leading to hidden financial losses.

This study was conducted to establish a data-driven understanding of production costs, profitability, and financial sustainability. Primary data covering April to October 2025 was collected from internal operational records, including sales quantities, selling prices, raw material consumption (Bana, Jari, Taani), labour wages, electricity expenses, and weekly colour/design-wise sales data. The analysis was performed using Microsoft Excel for cost accounting and break-even evaluation, while time series forecasting (ARIMA and STL decomposition) was applied to demand data.

The cost and profitability analysis revealed a major structural issue: although Lachha sarees accounted for approximately 69% of total sales volume, they consistently generated negative contribution margins due to under-pricing. Each Lachha saree sold increased total losses. In contrast, Aara sarees—despite representing only 31% of sales volume—maintained positive margins and achieved break-even at operationally feasible production levels. The break-even analysis confirmed that Aara sarees require only 26 units per month to cover fixed costs, whereas Lachha sarees have no achievable break-even point under current pricing. The minimum sustainable price for Lachha sarees was identified as ₹441 per unit.

Production estimation using the limiting material approach identified Taani as the primary production constraint. Insufficient inventory planning for Taani restricts output, leading to underutilization of labour and inflated per-unit costs. The absence of structured cost allocation further masked design-wise profitability differences.

Time series forecasting provided additional strategic insight. Weekly colour-based analysis showed that light shade sarees account for over 92% of sales but exhibit a declining trend over the study period. Dark shades, although currently representing a small share, show emerging growth potential. Design-wise forecasting for Aara sarees using an ARIMA(1,0,2) model demonstrated statistically reliable demand patterns, supporting scientific inventory and production planning.

The combined findings confirm that the core business problem lies in the absence of cost-based pricing, margin-focused production planning, and systematic monitoring of contribution margins. Without corrective action, increasing production of loss-making designs will worsen financial performance.

The study recommends immediate implementation of cost-based pricing for Lachha sarees, introduction of a monthly cost and margin tracking system, improved raw material planning specially for Taani and a strategic shift from volume-driven to contribution-driven production decisions. Adoption of demand forecasting models for design and colour planning will further enhance inventory optimization and reduce stock inefficiencies. Open a small or medium shop in the front portion of the house which definitely fruitful if followed with in time.

If implemented effectively, these recommendations can eliminate hidden losses, improve pricing accuracy, enhance capacity utilization, and transform Vishwas Maurya Textile into a financially sustainable, data-driven manufacturing operation.

2. Detailed Explanation of Analysis Process/Method

2.1 Data Cleaning and Preprocessing

After completing data collection, a systematic data cleaning and preprocessing process was undertaken to ensure accuracy, consistency, and reliability of the dataset before conducting cost analysis and forecasting. Since the data was sourced from handwritten operational book of the powerloom unit, it contained inconsistencies in recording formats, duplicate entries, missing values, and non-standardized measurement units. Therefore, the first step involved carefully reviewing sales, raw material consumption, and expense records to identify and remove duplicate entries to prevent double counting of quantities and revenue. Each transaction was cross-verified to ensure that totals reflected actual business activity.

Missing and incomplete values were then examined. Where daily records were having gaps but weekly, monthly totals figured out by aggregation methods, which used to reconstruct accurate totals. Date columns were converted into proper datetime format and arranged in chronological order to maintain time series structure. Some certain values logically derived such as weighted averages from each material consumption per saree which is additional information provided by owner. However, records with incomplete or non-recoverable information were excluded to maintain analytical integrity and avoid biased results.

To maintain uniformity across the dataset, all units of measurement were standardized. Raw materials including Bana, Jari, and Taani were converted into kilograms, and all monetary values were recorded in Indian Rupees (₹). Text-formatted numerical values were corrected to numeric format to prevent formula errors during computation.

For analytical clarity, aggregation reduced operational noise and allowed clearer identification of cost trends, while weekly structuring ensured equally spaced time intervals required for time-series models.

A key preprocessing applied on excess raw material consumption representing unsold production; therefore, total material costs were allocated across both sold and unsold sarees to ensure accurate per-unit cost computation and realistic inventory valuation. Since formal production records were unavailable, production quantity was estimated using the limiting material approach, where feasible output was determined by the minimum production supported by available raw materials.

Before proceeding to analysis, validation checks were performed to ensure internal consistency. Raw material consumption totals were reconciled with estimated production quantities, cost allocations were cross-verified across sheets, and negative or illogical values were eliminated. September and October data were excluded from detailed profitability calculations due to incomplete production records and shown value in excel is derived from average of other months that would otherwise mislead per-unit cost and margin results.

Stationarity was tested using the Augmented Dickey-Fuller (ADF) test. Finally, the cleaned and structured dataset was validated for consistency before applying STL decomposition and ARIMA modelling.

1.2 Comprehensive Explanation of Methods and Analysis

2.2.1 The lack of clarity regarding profitability and break-even performance per saree.

To structurally identify the issue, multi-stage analysis is implemented using descriptive statistics, cost accounting principles, constrained production estimation, profitability analysis, and break-even analysis with Microsoft Excel, which serves as the primary analytical tool due to its accessibility, transparency, and suitability for small-scale business decision-making.

Stage 1: Descriptive Analysis of Sales, Production, and Raw Material Data

The first stage involved descriptive analysis, aimed at developing an overall understanding of historical business performance. Raw data on saree sales, selling prices, and raw material consumption were aggregated on a monthly basis to reduce daily-level noise and reveal meaningful trends over time.

Mathematically, month-wise aggregation was expressed as:

$$Q_m = \sum_{i=1}^n q_i$$

where

Q_m = total quantity in month m

q_i = individual transaction quantity

Pivot Tables and summary statistics (mean, total, and proportion) were used to analyse variability in demand and pricing across Aara and Lachha saree designs. This method was chosen because descriptive analysis provides a non-assumptive foundation, allowing patterns to emerge directly from the data rather than imposing theoretical constraints. It directly supported the problem statement by highlighting inconsistencies in sales volume and pricing that warranted deeper cost investigation.

Stage 2: Production Estimation Using Limiting Material Approach

A major analytical challenge was the absence of formal production records. To overcome this, a limiting material approach was adopted to estimate realistic production quantities. This approach is grounded in operational theory, where output is constrained by the scarcest input resource.

First, weighted average raw material consumption per saree was calculated based on the product mix:

$$\bar{C}_r = \sum_{j=1}^k w_j \cdot c_{rj}$$

where

$\bar{C}_r$ = average consumption of raw material r per saree

w_j = proportion of design j in total production

c_{rj} = consumption of raw material r for design j

Next, total production supported by each raw material was computed:

$$P_r = \frac{T_r}{\bar{C}_r}$$

where

T_r = total quantity of raw material r available

The actual production quantity was determined as:

$$P = \min (P_{\text{Bana}}, P_{\text{Jari}}, P_{\text{Taani}})$$

This method was selected because it ensures feasible and conservative production estimates, preventing overstatement of output and underestimation of cost per unit—both critical to accurate profitability analysis.

Stage 3: Cost Structure and Cost Allocation Analysis

The third stage focused on building a transparent cost structure aligned with cost accounting principles. Costs were classified as:

Variable Costs: Raw materials (Bana, Jari, Taani) and labour wages

Fixed Costs: Electricity expenses

Total raw material cost was computed as:

$$\text{Material Cost} = \sum_{r=1}^3 Q_r \times CP_r$$

Labour cost per saree was allocated as:

$$\text{Labour Cost per Saree} = \frac{\text{Total Monthly Wages}}{P}$$

Fixed electricity cost was apportioned per saree unit:

$$\text{Electricity Cost per Saree} = \frac{\text{Total Electricity Cost}}{P}$$

A key assumption allocating material costs across both sold and unsold sarees was applied to reflect inventory reality and prevent artificial inflation of per-unit profit. Separate cost sheets for Aara and Lachha sarees enabled product-wise cost visibility, which directly addressed the problem of unclear design-level profitability.

Stage 4: Profitability Analysis

Profitability per saree was computed as:

$$\text{Profit per Saree} = \text{Selling Price} - \text{Total Cost per Saree}$$

Percentage margin was calculated to enable comparability across designs and months:

$$\text{Profit Margin (\%)} = \frac{\text{Profit}}{\text{Selling Price}} \times 100$$

This analysis revealed pricing inefficiencies, particularly instances where selling price failed to cover full cost—clearly linking back to the original problem of poor cost visibility.

Stage 5: Break-Even Analysis

Finally, break-even analysis was conducted to determine financial sustainability. Contribution per saree was calculated as:

$$\text{Revenue} = \text{Selling Price} \cdot \text{Quantity}$$

$$\text{Total Cost} = \text{Fixed Cost} + \text{Variable Cost} \cdot \text{Quantity}$$

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

$$\text{Contribution} = \text{Selling Price} - \text{Variable Cost}$$

Break-even occurs when:

$$\text{Revenue} - \text{Total Cost} = 0$$

So:

$$\text{Contribution} * \text{Quantity} - \text{Fixed Cost} = 0$$

Break-even quantity was derived using:

$$\text{Break-even Quantity} = \frac{\text{Fixed Cost}}{\text{Contribution}}$$

Break-even exists if and only if:

$$C > 0$$

(Contribution per unit is greater than zero.)

i.e.,

$$P > V$$

(Selling Price per unit is greater than Variable Cost per unit.)

2.2.2 Lack of understanding in sales forecasting of Designs/Colour

The forecasting framework is diploid to support analytical demand planning. The process combined decomposition techniques, stationarity testing, autocorrelation diagnostics, ARIMA modelling, model validation, and forecast generation.

Stage 1: Time-Series Framework and Data Structure

The forecasting analysis used weekly aggregated sales data over 28-30 observations depending on category (design-wise and colour-wise). Weekly frequency was selected because it balances with stability, captures short-term demand shifts and provides sufficient observations for ARIMA modelling.

Let:

$$Y_t = \text{Sales quantity at time } t$$

Where $t = 1, 2, 3, \dots, n$.

The goal of forecasting is to estimate:

$$\hat{Y}_{t+h}$$

Where h is the forecast horizon (next 10 weeks).

Stage 2: Exploratory Time-Series Decomposition (STL Method)

Before fitting parametric models, STL (Seasonal-Trend decomposition using Loess) was applied to decompose the series into:

$$Y_t = T_t + S_t + R_t$$

Where:

T_t = Trend component

S_t = Seasonal component

R_t = Residual (random noise)

Purpose of Decomposition is to Identify long-term direction along with to detect repeating weekly seasonal patterns and Understand volatility magnitude.

This step prevented incorrect model specification by separating structural patterns from noise.

Stage 3: Stationarity Testing

ARIMA models require stationarity (constant mean and variance over time) which was tested using the Augmented Dickey-Fuller (ADF) Test.

ADF Hypothesis:

H_0 :- Series has a unit root (non-stationary)

H_1 :- Series is stationary

If required, differencing would be applied:

$$Y'_t = Y_t - Y_{t-1}$$

Stage 4: Autocorrelation Analysis (ACF and PACF)

To determine model structure:

ACF (Autocorrelation Function) :- identifies moving average order (q).

PACF (Partial Autocorrelation Function) :- identifies autoregressive order (p).

Stage 5: ARIMA Model Specification

The ARIMA(1,0,2) model is mathematically expressed as:

$$Y_t = c + \phi_1 Y_{t-1} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \varepsilon_t$$

Where:

c = constant term (baseline demand level)

ϕ_1 = autoregressive coefficient (captures demand persistence)

θ_1, θ_2 = moving average coefficients (capture short-term shocks and adjustment effects.)

ε_t = white noise error term

Stage 6: Model Selection Criteria

Model comparison was conducted using:

AIC (Akaike Information Criterion):- This measure evaluates how well the model fits the data while penalizing extra complexity.

BIC (Bayesian Information Criterion):- It applies a stronger penalty for adding too many parameters. This helps avoid overfitting, especially when the dataset is small.

HQIC (Hannan-Quinn Information Criterion):- This criterion also balances model fit and complexity, with a penalty strength between AIC and BIC.

Lower values indicate better model fit with optimal complexity in all three.

Stage 7: Model Diagnostics and Validation

To ensure reliability:

Ljung-Box Test :- It's checks whether the model's residuals are independent over.

Jarque-Bera Test :- It's Checks whether the residuals follow a normal distribution.

Heteroskedasticity Test :- Checks whether the variance of residuals remains constant over time (no changing volatility).

Stage 8: Forecast Generation

Using estimated parameters, forecasts were generated recursively:

$$\hat{Y}_{t+h} = c + \phi_1 \hat{Y}_{t+h-1} + \theta_1 \hat{\varepsilon}_{t+h-1}$$

$\hat{Y}_{t+h}$ = Forecasted value at time $t + h$

c = Constant (intercept term)

ϕ_1 = Autoregressive parameter of order 1

θ_1, θ_2 = Moving average parameters

$\hat{\varepsilon}_{t+h-1}$ = Forecasted residuals

2.3 Justification of Tools and Methods

Microsoft Excel was selected due to its flexibility, auditability, and alignment with small-business operational capabilities. The chosen methods collectively ensured logical cost allocation, realistic production estimation, and actionable financial insight, transforming fragmented operational data into a coherent, decision-ready analytical framework.

ARIMA modelling was chosen because it captures temporal dependence in demand which does not require external explanatory variables. It performs well with moderate sample sizes and widely accepted in business forecasting. STL decomposition was used to understand structural components before parametric modelling. The approach balances statistical rigor with practical business applicability.

3. Results and Findings

3.1 Design-wise Sales Contribution and Product Mix

The sales ratio chart derived from April to October data shows that Lachha sarees account for approximately 69% of total sales, while Aara sarees contribute around 31%. This clearly indicates that Lachha sarees dominate the product mix in terms of volume. However, higher sales volume does not automatically translate into profitability, which becomes evident in subsequent cost and margin analysis.

This imbalance suggests that production planning is demand-driven rather than margin-driven. The business prioritizes Lachha sarees due to higher sales frequency, but without a structured cost framework, this strategy exposes the business to sustained losses despite strong market demand.

3.2 Raw Material Consumption and Production Capacity Analysis

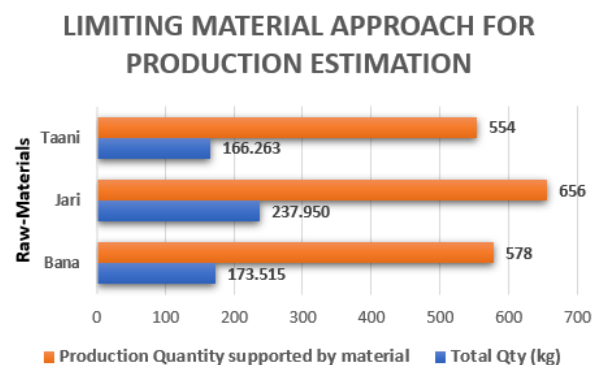


Figure 1: Limiting Material Approach

The material consumption analysis, supported by summary charts (Figure 1), reveals that Taani is the limiting raw material. While Bana and Jari could support higher production volumes, Taani availability restricts overall output. Using the limiting material approach, the final feasible production quantity was estimated at 554 sarees.

This finding is critical because it highlights that production is constrained not by labour or machinery, but by material planning. The absence of structured inventory monitoring for Taani leads to underutilization of machines and labour, directly affecting cost efficiency.

3.3 Cost Structure Analysis (Aara and Lachha Sarees)

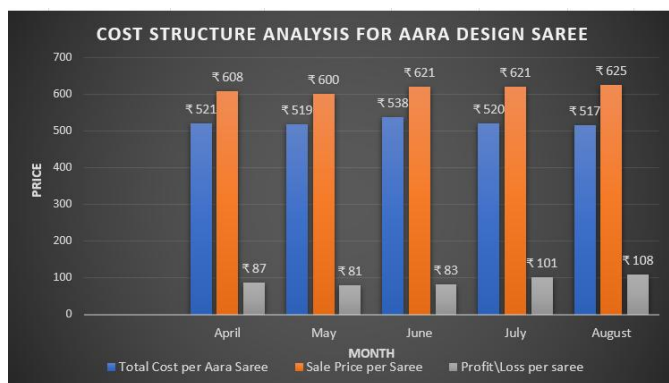


Figure 2: Cost Structure Analysis for Aara Saree

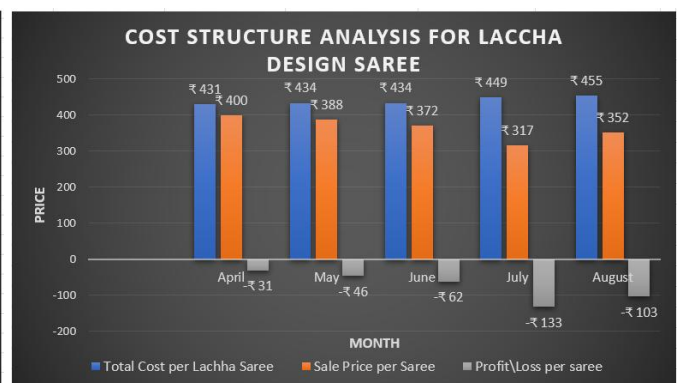


Figure 3: Cost Structure Analysis for Laccha Saree

Bar charts comparing cost components show that raw material costs and wages together form the largest portion of total production cost. For Aara sarees, the average total cost per unit ranged between ₹517–₹540 (Figure 2), whereas for Lachha sarees, the cost ranged between ₹431–₹455 (Figure 3).

Although Lachha sarees appear cheaper to produce, the selling price trend reveals a critical issue. Aara sarees consistently sell above cost.

3.4 Month-wise Profit and Loss Trends

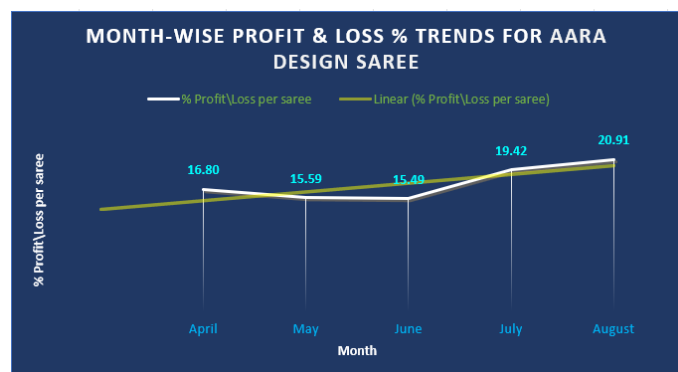


Figure 4: Month-wise Profit & Loss % Trends for Aara Saree

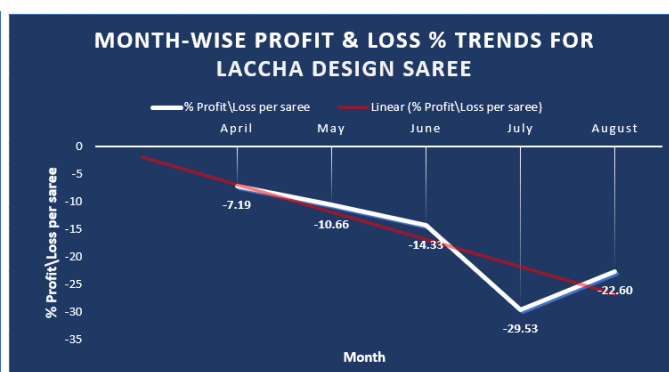


Figure 5: Month-wise Profit & Loss % Trends for Laccha Saree

The monthly profit and loss charts highlight that Aara sarees maintain stable profitability even during fluctuating production months, generating positive margins between 11% and 21% (Figure 4). The upward trend line in profit percentage for Aara sarees indicates improved pricing efficiency over time.

Conversely, the Lachha saree profit trend line consistently remains below zero, with losses ranging from 7% to nearly 30% per saree (Figure 5). Even during months with higher sales volume (e.g., July and August), losses increased due to rising material costs and unchanged selling prices. This contrast exposes a major flaw in pricing strategy such as high-volume products are being under-priced, while lower-volume products are subsidizing losses. This clearly demonstrates that increasing sales volume without correcting price or cost structure amplifies losses.

September and October were excluded from detailed interpretation due to zero or incomplete production data, resulting in division errors. Including these months would distort trend interpretation and reduce analytical reliability.

3.5 Break-Even Analysis Results

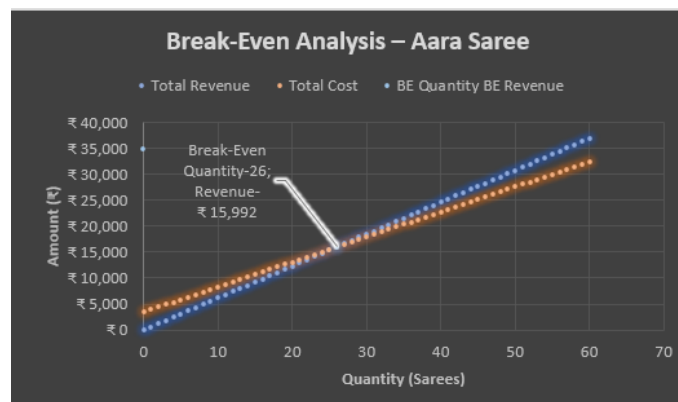


Figure 6: Break-Even Analysis for Aara Saree

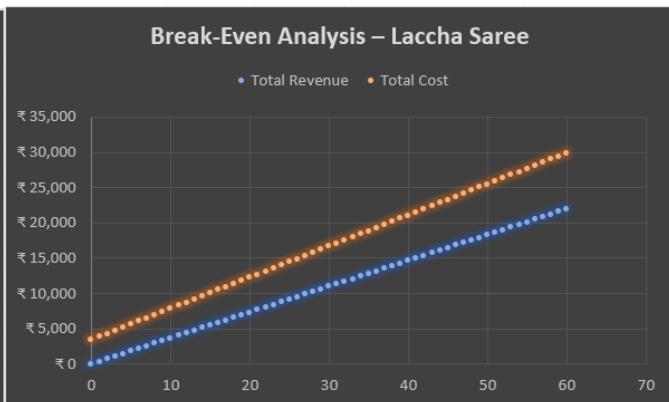


Figure 7: Break-Even Analysis for Laccha Saree

The break-even analysis chart and tables provide the most direct insight into financial sustainability. Aara sarees show a positive contribution of ₹132 per unit, resulting in a break-even quantity of 26 sarees per month (Figure 6), which is operationally achievable.

Lachha sarees show a negative contribution of ₹75 per unit, indicating no break-even point (Figure 7) at current pricing.

This finding is crucial because every Lachha saree sold increases total loss, regardless of volume. The minimum price required for Lachha sarees to avoid losses is ₹441, significantly higher than the current average selling price.

3.6 Time Series Decomposition of Weekly Sales Data based on Designs

3.6.1 STL analysis of Aara Design Saree

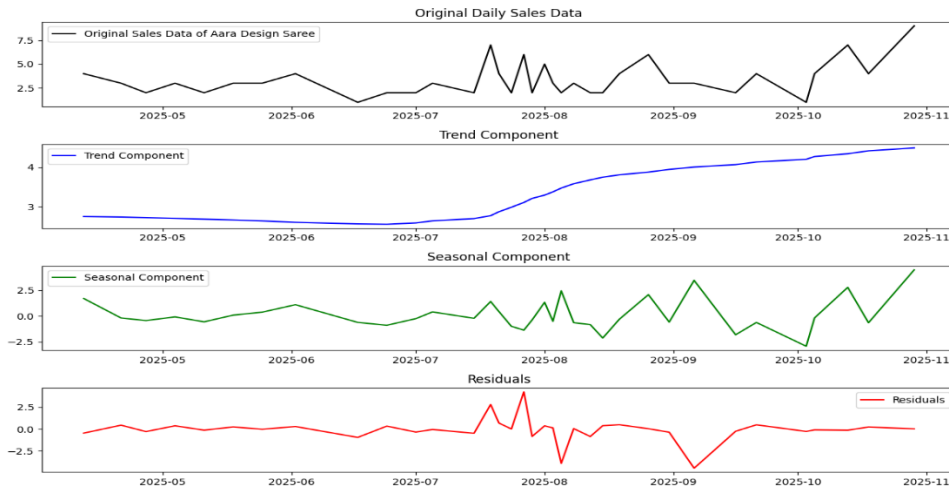


Figure 8: Time Series Decomposition of Weekly Sales for Aara Saree

The time series decomposition of Aara daily sales shows that the trend component gradually increases around 3 sarees per day in April to about 5 sarees per day by late October, indicating steady demand growth. The seasonal component fluctuates within a range of roughly -2.5 to $+3$ sarees (Figure 8), suggesting moderate short-term variations but no strong recurring seasonal cycle. The original sales values vary between 1 and 9 sarees per day, reflecting noticeable volatility. Residuals mostly lie between -4 and $+4$ (Figure 8), centered around zero, which indicates that most systematic variation is explained by the trend and seasonal components. Overall, Aara sales exhibit gradual growth with moderate randomness and limited seasonality.

3.6.2 Model Performance and Statistical Validation

The analysis tested different ARIMA models to find the best way to forecast Aara Design Saree quantities. The ARIMA(1,0,2) model was found to be the best choice. It performed better than other models through lower information criteria values and excellent diagnostic test results. With an AIC of 145.422, the preferred model fits the data better and is more efficient. The Ljung-Box test shows a probability of 0.71, which means there is no significant autocorrelation in the residuals. This confirms that the model captures demand patterns effectively. The Jarque-Bera test shows a probability of 0.66, which means the residuals follow a normal distribution. This is important for making reliable forecasts.

The constant term coefficient of 3.5493 with a p-value of 0.000 shows there is a significant baseline demand level for Aara Design Sarees. The moving average terms, especially the second lag (ma.L2) with a coefficient of 0.6266 and p-value of 0.001, are statistically significant. This shows the model captures recent demand changes and adjustments well which means changes have a large influence on future demand patterns, making the model effective for understanding supply and demand dynamics.

The heteroskedasticity test with a probability of 0.05 shows that variance is fairly consistent across the time series, which is acceptable for forecasting. However, it suggests some minor volatility clustering in demand patterns. The skewness value of 0.37 and kurtosis of 2.89 show that the residuals are nearly normally distributed, with a slight positive skew indicating occasional demand spikes. All these statistical measures together confirm that the ARIMA(1,0,2) model successfully captures how demand changes over time and the randomness in demand.

The diagnostic test results support choosing this model. ARIMA(1,0,2) shows a Ljung-Box probability of 0.71, which means the residuals are clean and there is no remaining autocorrelation structure. The Jarque-

Bera test strengthens this conclusion. ARIMA(1,0,2) achieves a probability of 0.66 (showing good normal distribution). All these test results clearly show that ARIMA(1,0,2) is balances model with prediction accuracy.

The autoregressive coefficient of 0.3319 shows moderate persistence in demand. This means that demand shocks gradually disappear over time rather than immediately going back to normal levels.

The sigma-squared value of 2.7211 represents the part of demand that the model cannot explain. This represents the natural unpredictability in customer demand for this specialized product. Together, these insights show that Aara Design Sarees have predictable demand patterns.

3.6.3 Forecast Analysis

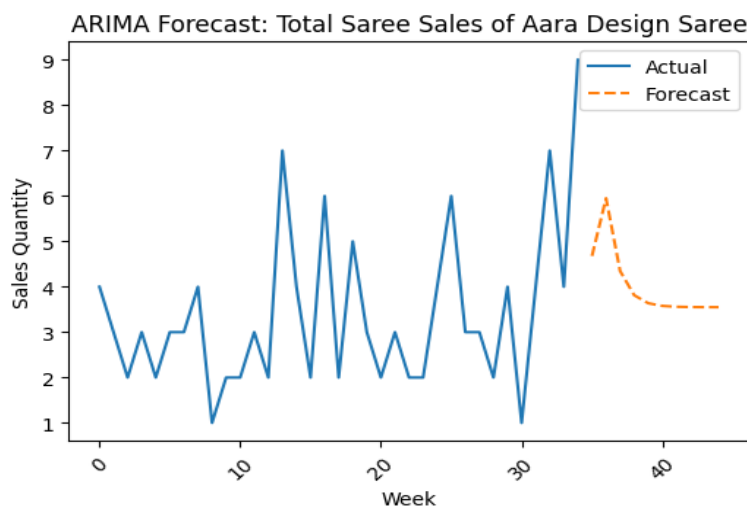


Figure 9: Weekly Forecast Graph using ARIMA(1,0,2) for Aara Saree

The ARIMA forecast for Aara design saree sales indicates that weekly demand fluctuates moderately but does not show a strong upward or downward trend. After a short-term spike, the forecast stabilizes around 3–4 sarees per week (Figure 9), suggesting steady but limited demand growth. This implies that Aara sales are relatively stable and primarily demand-driven rather than trend-driven. From a financial perspective, the design generates consistent but modest profit per unit. Therefore, production planning should align closely with forecasted demand to avoid excess inventory and unnecessary cost absorption.

3.6.4 STL analysis of Laccha Buti Design Saree

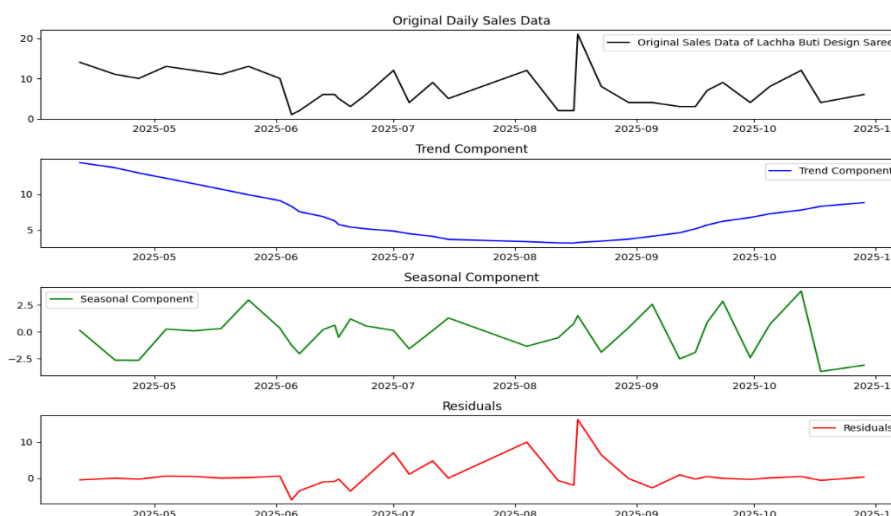


Figure 10: Time Series Decomposition of Weekly Sales of Laccha Buti Saree

The decomposition of Lachha Buti daily sales shows that the trend component declines significantly from approximately 12-15 sarees per day in April-May to around 3–4 sarees per day by August, before recovering to nearly 8 sarees per day in October(Figure 10), indicating a mid-period demand slowdown followed by improvement. The original sales fluctuate widely between 2 and 20 sarees per day, showing much higher volatility compared to Aara. The seasonal component varies roughly between -3 and $+3$ sarees, suggesting moderate short-term cyclical movement but no strong fixed seasonal structure. Residuals range approximately from -3 to $+10$, with a sharp positive spike around late August, indicating occasional abnormal demand shocks. Overall, Lachha sales exhibit higher volatility, a temporary demand contraction mid-period, and stronger irregular fluctuations than Aara.

3.6.5 Model Performance and Statistical Validation

The ARIMA(2,0,1) model was estimated using 35 daily observations to forecast Lachha Buti Design Saree demand. Model selection was guided by information criteria, with $AIC = 148.01$ and $BIC = 155.78$. These criteria balance goodness of fit with model simplicity, ensuring the model is neither overfitted nor unnecessarily complex. Overall, the selected specification provides a reasonable statistical fit to the observed data.

The Ljung-Box test produced a high probability value ($p = 0.97$), indicating no significant autocorrelation in the residuals. This confirms that the model successfully captures the serial dependence structure in daily demand. However, the Jarque-Bera test ($p = 0.02$) suggests that residuals are not normally distributed, implying the presence of occasional extreme demand fluctuations.

Further analysis shows positive skewness and slightly higher kurtosis, indicating demand spikes and heavier tails than a normal distribution. The heteroskedasticity test ($p = 0.02$) reveals non-constant variance over time, suggesting periods of higher demand volatility, possibly driven by seasonal or promotional effects.

3.6.6 Forecast Analysis

The constant term is statistically significant, representing a stable baseline demand level. In contrast, most AR and MA coefficients show limited statistical strength, suggesting weak persistence of short-term shocks. Overall, the model captures demand patterns adequately, though volatility remains present.

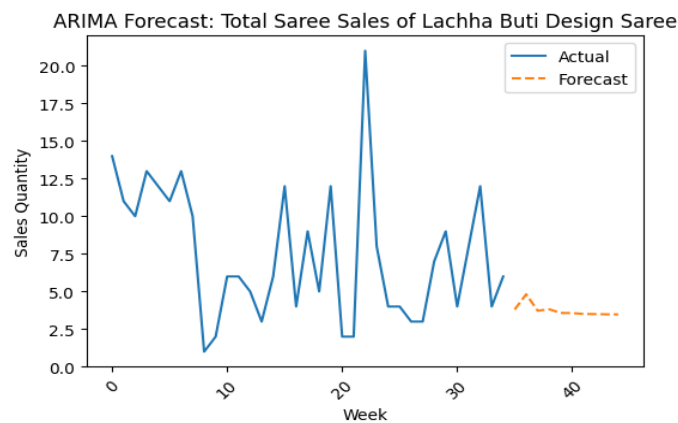


Figure 11: Weekly Forecast Graph using ARIMA(2,0,1) for Laccha Buti Saree

The ARIMA forecast for Lachha Buti saree sales show a stabilization of demand in the upcoming weeks. forecasted values gradually converge toward a steady range of approximately more than 3 to 5 units per week (Figure 11), which is normalization around the long-run mean. The declining forecast variance implies that extreme fluctuations observed in past weeks are unlikely to persist in the short term. Overall, the model predicts moderate and stable future demand, supporting controlled production planning with limited safety stock adjustments.

3.7. Saree Colour Shade Sales Analysis

3.7.1 STL analysis of Saree as per Colour Shade

The STL analysis breaks down the sales data into different parts: the main trend, regular patterns, and random changes. For light shades, the trend component shows an initial decline early mid-year followed by stabilization around mid-year and gradual recovery, indicating evolving baseline demand over time. The seasonal component highlights consistent weekly fluctuations, suggesting predictable cyclical purchasing behaviour. The changes usually range from -5 to +6 units around the average trend. The residual component captures occasional demand spikes and short-term volatility that are not explained by trend or seasonality. Together, these components confirm that demand is moderately systematic which suitable to use of time-series forecasting.

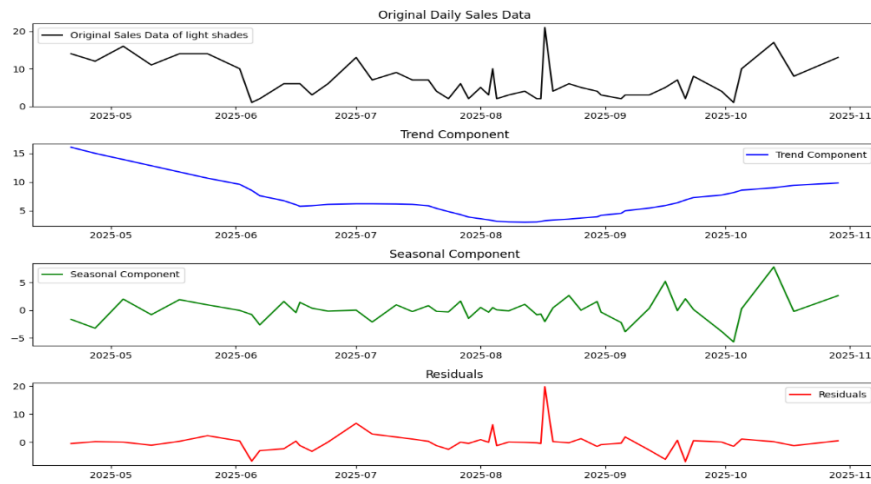


Figure 12: Time Series Decomposition of Weekly Sales for Light Shade Saree

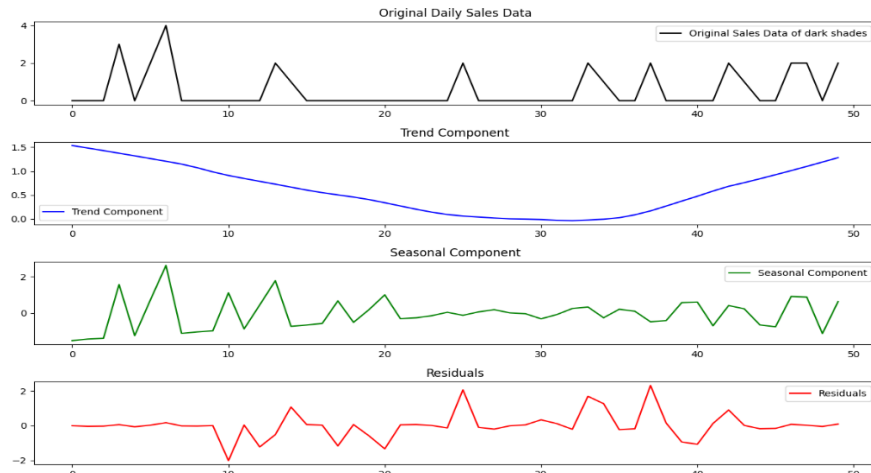


Figure 13: Time Series Decomposition of Weekly Sales for Dark Shade

Whereas for dark shade saree sales shows that the trend component declines from approximately 1.5 units at the beginning to nearly 0.2 units mid-period, before recovering to around 1.3 units (Figure 12), indicating weak but slightly improving baseline demand. The seasonal component fluctuates roughly between -1.0 and +2.5 units (Figure 12), showing mild weekly variation with limited cyclical strength. Compared to light shades, the seasonal amplitude is relatively small, confirming lower structured demand patterns. The residual component ranges approximately between -2 and +2 units (Figure 13), highlighting occasional random spikes and drops. Overall, dark shade demand remains low (generally 0–2 units daily) with higher volatility and limited persistence, supporting a cautious production approach.

3.7.2 Model Performance and Statistical Validation

Stationarity was tested using the Augmented Dickey-Fuller (ADF) test. For light shades, the ADF statistic was -2.947 with a p-value of 0.0436, leading to rejection of the unit root hypothesis and confirming overall stationarity. However, minor fluctuations suggest the presence of slight non-stationary elements. For dark shades, the ADF statistic was approximately -3.107, indicating comparatively stronger stationarity. ACF and PACF plots showed that light shades have a strong relationship between current sales and sales from the previous two weeks. Dark shades displayed a similar but more volatile pattern. These observations supported selection of the ARIMA(1,0,2) model.

The ARIMA(1,0,2) model for light shades produced a significant constant term (7.0779), along with $AR(1) = 0.6742$, $MA(1) = -0.4236$, and $MA(2) = 0.0862$. Model fit indicators were $AIC = 299.348$ and $BIC = 308.908$. Diagnostic tests showed no residual autocorrelation, confirming proper model specification. However, the Jarque-Bera test indicated non-normal residuals, suggesting occasional outliers, while the heteroskedasticity test confirmed stable variance.

For dark shades, parameter estimates were broadly similar, and the Ljung-Box test confirmed good fit. Nevertheless, non-normal residual patterns suggest forecasts should be interpreted as short-term directional guidance rather than exact predictions.

3.7.3 Forecast Analysis

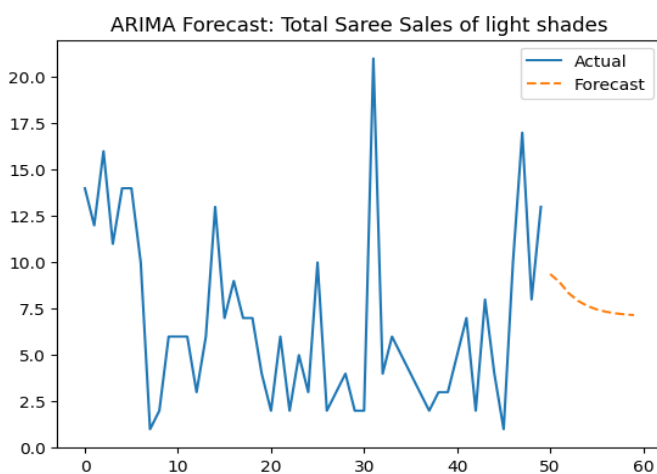


Figure 14: Weekly Forecast using ARIMA(1,0,2) for Light shades

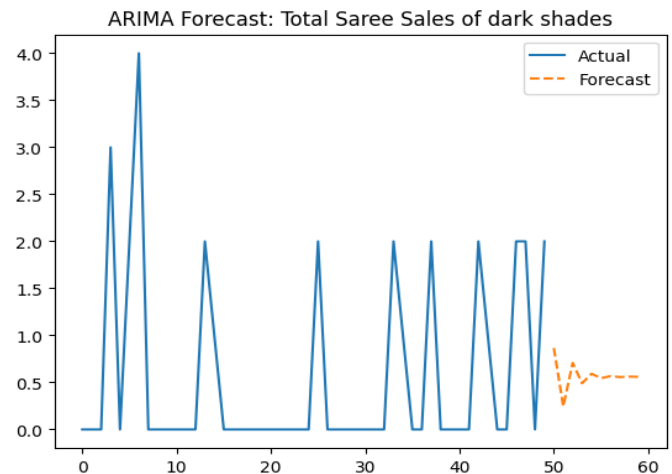


Figure 15: Weekly Forecast using ARIMA(1,0,2) for Dark shades

The model predicts future sales for the next 10 weeks. For light shades, the forecast shows sales will level off around 8-11 units per week (Figure 14). This is better than the low point in September (4.875 units), but still much lower than the peak in April-May (13-15 units). This indication of the market has changed rather than just having a temporary drop.

For dark shades, the forecast predicts sales will slowly increase to around 1-2 units per week (Figure 15). Currently, dark shades average 0.56 units per week, so this would be about 2-3 times higher. This suggests that dark shades might be popular and demand surge around winter, wedding occasion and year-end festivals with customers.

4. Interpretation of Results and Recommendations

4.1 Interpretation of Results

4.1.1 Interpretation for Production Cost and Profitability

The results of the analysis directly address the core problem statement: the absence of structured production cost tracking and break-even awareness has led to unclear profitability and hidden losses within Vishwas Maurya Textile.

The findings reveal that while Lachha sarees dominate sales volume, they consistently generate negative contribution margins, meaning that higher sales increase total losses rather than profits. This indicates a critical disconnect between market demand and cost-based pricing. In contrast, Aara sarees, despite lower sales volume, remain consistently profitable, demonstrating that profitability is driven by margin efficiency rather than quantity sold.

The production analysis further highlights that output is constrained by Taani availability, not by labour or machinery. This implies that operational inefficiencies stem from poor raw material planning rather than capacity limitations. As a result, labour and fixed costs are spread over fewer units, inflating per-unit cost and reducing competitiveness.

Cost structure and break-even analysis show that Lachha sarees currently have no achievable break-even point, which poses a serious threat to financial sustainability. Without corrective action, continued production of Lachha sarees at current prices will erode overall profitability, even if sales volumes increase.

Overall, the analysis confirms that the business lacks cost visibility, margin-based pricing, and informed production planning, making it vulnerable to sustained losses despite stable demand.

Light shade forecasting is reliable for short-term decisions. Dark shade forecasting provides directional insight but should not drive major production expansion without further observation.

4.1.2 Interpretation for Aara, Lachha Buti Design & Colour Shades Sarees

The analysis of weekly sales data in different dimension of Aara, Lachha Buti, and colour shade categories. Aara Design Sarees exhibit steady and predictable demand growth, with the trend gradually increasing from around 3 to 5 units over the study period. Seasonal effects are moderate and residuals remain well-behaved, confirming that the ARIMA(1,0,2) model captures demand dynamics effectively. Diagnostic tests (high Ljung-Box and Jarque-Bera probabilities) indicate clean residual structure and normality, suggesting reliable short-term forecasting. Business-wise, Aara generates stable and consistent demand, making it suitable for planned production with controlled inventory levels.

In contrast, Lachha Buti Design Sarees show higher volatility and a noticeable mid-period demand decline before recovery. Although the ARIMA(2,0,1) model removes autocorrelation effectively (high Ljung-Box p-value), residual non-normality and heteroskedasticity indicate occasional demand spikes and variance instability. This means forecasts are directionally reliable but subject to sudden fluctuations. Production planning should therefore include moderate safety stock to manage volatility risks.

Regarding colour shades, light shades demonstrate stronger structured demand with meaningful seasonal cycles and moderate persistence, supporting reliable ARIMA-based forecasting. Dark shades, however, maintain low baseline demand with weaker seasonality and higher relative volatility. Forecasts suggest light shades will stabilize at moderate levels, while dark shades may see slight improvement but remain limited in volume. Overall, Aara and light shades offer predictable and stable demand patterns, whereas Lachha Buti and dark shades require more cautious inventory and demand management due to higher variability.

4.2 Actionable Recommendations (SMART)

Recommendation 1: Implement Cost-Based Pricing for Lachha Sarees (Urgent)

The selling price of Lachha sarees should be revised to at least ₹441 per unit to eliminate negative contribution, which achieve non-negative contribution within one month then Price increase aligns with observed cost structure. Production decisions should prioritize sarees with positive contribution margins, particularly Aara sarees, until Lachha pricing is corrected. Improve overall monthly profit margin by 10–15%.

Recommendation 2: Introduce Monthly Cost and Margin Tracking System (Urgent)

By following provided Cost and Profitability Sheet or a standardized Excel-based cost sheet should be maintained monthly to track material cost, labour cost, total cost, selling price, and profit margin per saree design which Measure Monthly profit/loss through which visibility achieved by using existing Excel capability which prevents hidden losses, Start from the next starting of month or production cycle.

Recommendation 3: Improve Taani and Other Raw-Material Planning (Short-Term)

Minimum stock levels for Taani with proportional with other raw-materials should be maintain by Implement within 2 months, which defined to prevent production bottlenecks and underutilization of labour by following: Set monthly minimum Taani requirement based on average consumption.

Recommendation 4: Saree Colour Shade Strategy (Short-Term)

Start paying attention to dark shades because they are growing. Look at when is buying happening of dark shades. Think about adding more dark shade options and promoting them more. Plan to keep light shade inventory at around 9-11 units per week based on the forecast.

Recommendation 5: Dealer Strategy (Medium-Term)

Classify dealers into Stable demand dealers (regular weekly orders) and Volatile demand dealers (irregular large orders). Provide Priority production allocation to stable dealers and some bulk order discount or controlled credit terms for volatile dealers.

Recommendation 6: Sarees Stock Maintenance (Medium-Term)

Set inventory targets as 1.5 times the forecasted demand quantity to protect against running out of stock while managing carrying costs. Expected benefit: 15-20% reduction in stockouts while maintaining current inventory investment levels.

Recommendation 7: Add External Factors to Forecasting Models (Medium-Term)

The current ARIMA model relies entirely on demand history, missing opportunities to integrate forward-looking information about market conditions.

Start also looking at outside market factors such as Festival and wedding season months, Local fashion trends and colour demand, Competitor pricing in nearby markets, Dealer feedback about customer preference and Any promotion or discount activity.

Recommendation 8: Use Highway Opportunity for Direct Sales (Long-Term)

Since a new highway has been built in front of the house, this is a big opportunity. Open a small or medium shop in the front portion of the house within the next 3–6 months. Keep it simple but clean and well-arranged. Display best-selling and attractive sarees in the front so passing customers can easily see them. Install visible signboard facing highway. Offer small festival or bundle discounts.

4.3. Impact and Benefits of Implementation

Implementing these recommendations will transform Vishwas Maurya Textile from a volume-driven operation into a cost-aware, margin-focused business. Overall, the scenario is, pricing and cost controls will immediately stop recurring losses, while improved material planning will enhance capacity utilization and reduce per-unit cost which leads to stabilize cash flow, improve profitability, and enable data-driven decision-making.

As per designs and colours forecasting, production will become more planned and demand-driven. Less unsold stock and better use of raw materials.

By utilizing of available opportunities of newly build Highway by opening a shop on highway side can increase direct customer sales and reduce full dependence on dealers. Since no middleman margin is involved, profit per saree can increase at least by 10–15%. It will also Improve cash flow (instant payment), Create regular local customer base and Make demand more stable and predictable.

Links:

Spreadsheets Folder Link : [

<https://drive.google.com/drive/folders/1URzwylhewnk2dPJ7bMQXDwohv-pAZZXq?usp=sharing>]

Collab Folder Link:- [

https://drive.google.com/drive/folders/1Ti801BLa_ey2_vME0WQMJMFMadXUGtUd?usp=sharing]

Additional Data Link:- [

<https://drive.google.com/drive/folders/1nBVOxDj3kuZwLsaxjnSJMtqNILpBJALj?usp=sharing>]